

End Semester - Winter Examination (2024-2025)

VIII Semester B.Tech. (Mining Engineering)

Underground Space Technology (MND407)

Time: 2 Hours

Max. Marks: 50

Instructions: 1) Answer any FIVE questions 2) Assume suitable data wherever required
3) Illustrate your answers with neat schematic diagrams wherever required

Q. No	Question	Marks
1.	A long opening of circular cross section is located 1000 m below ground surface. In the plane perpendicular to the tunnel axis, the field principal stresses are vertical and horizontal. The vertical stress p is equal to the depth stress, and the horizontal stress is defined by $0.28p$. The unit weight of the rock mass is 27 kN/m^3 , the compressive strength is defined by a Coulomb criterion with cohesion, $c = 20 \text{ MPa}$, friction angle, $\phi = 25^\circ$ and the tensile strength by $\sigma_t = 0$. (i) Predict the response of the excavation peripheral rock to the given conditions. (ii) Propose an alternative design for the excavation.	10
2.	a) Explain briefly the procedure for monitoring the stability of large underground excavations in hard rock using geotechnical instrumentation. b) Estimate unsupported span and support pressure in a permanent rock cavern from the following data: Rating of $J_a = 2.0$, $J_r = 3.0$, $J_n = 1.0$, $\text{SRF} = 1.0$, $J_w = 1.0$ No. of joints present in a rock mass per cubic meter = $15/\text{m}^3$ and No. of joint sets = 5.	5 5
3.	Classify the Tunnel Boring Machines (TBM) based on their design and suitability for different ground conditions and state their applications and limitations.	10
4.	Briefly discuss the importance of ground response curves and explain the influence of support stiffness and of the timing of its installation on support performance in a rock cavern. Illustrate your answer with neat schematic diagrams.	10
5.	Describe the storage and design principles of rock caverns for underground storage of crude oil and LNG (Liquefied Natural Gas). Illustrate your answer with neat schematic diagrams.	10
6.	a) Compute Q , when; $J_n = 12$; $J_r = 1$, $J_a = 2$, $J_w = 0.66$, $\text{SRF} = 5$ and volumetric joint count = 33. Using this, answer the following (i) How the Q value would change in case of tunnel portals? (ii) What ground condition would you expect and provide the support recommendations to construct an 8 m diameter tunnel at a depth of 400 m. b) A rock mass has the following characteristics: the compressive strength of the intact rock is 160 MPa, the Q value of rock mass is 25 and the RMR (Rock Mass Rating) is 35%. Estimate the in-situ deformation modulus.	6 4

Note: Use the following formulas wherever required

$$\sigma_{rr} = \frac{p}{2} \left[(1+K) \left(1 - \frac{a^2}{r^2} \right) - (1-K) \left(1 - 4\frac{a^2}{r^2} + \frac{3a^4}{r^4} \right) \cos 2\theta \right]$$

$$\sigma_{\theta\theta} = \frac{p}{2} \left[(1+K) \left(1 + \frac{a^2}{r^2} \right) + (1-K) \left(1 + \frac{3a^4}{r^4} \right) \cos 2\theta \right]$$

$$\sigma_{r\theta} = \frac{p}{2} \left[(1-K) \left(1 + \frac{2a^2}{r^2} - \frac{3a^4}{r^4} \right) \sin 2\theta \right]$$